IE 3080: Probabilistic Analysis of Algorithms and Randomized Algorithms

Advanced Topics, Spring 2026

Course Syllabus

General Information

Meeting logistics: 01/15/26 – 04/23/26

Thursdays 6 PM – 8:30 PM in 1041 Benedum Hal

Instructor: Amin Rahimian

Course Rationale

Randomization is a powerful tool in algorithm design when deterministic solutions are inefficient or hard to come by. Probabilistic analysis provides performance guarantees that reflect input properties and problem uncertainty. This advanced topics course introduces fundamental tools (concentration bounds, coupling, random graphs, probabilistic method, martingales) and explores seminal areas (optimal stopping, prophet and secretary inequalities, Pandora's box, multi-armed bandits, and inference/learning on trees and networks). The course emphasizes both technical mastery and critical engagement with research papers.

Course Description

The course is organized into two parts. The first half focuses on probability and discrete analysis tools, including concentration bounds, convergence analysis, the probabilistic method, random graphs, and combinatorial techniques. The second half covers special topics and seminal papers in optimal stopping (prophet inequalities, secretary and Pandora's box problems), bandit algorithms, and inference & learning on trees and networks. Student work is evaluated via active attendance and homework problems in the first part and active attendance and research-paper presentations in the second part.

Learning Objectives

- Apply concentration inequalities, coupling, and random graph tools to analyze algorithm performance.
- Derive and interpret PAC learning bounds, VC-dimension, and Rademacher complexity.
- Understand and critique seminal results on prophet inequalities, secretary/Pandora's box, and multi-armed bandits.
- Analyze algorithms for inference and learning on trees and networks.
- Develop skills in presenting technical research and engaging in scholarly discussion.

Pre-Requisites / Co-Requisites

Strong background in probability theory assumed; prior exposure to design & analysis of algorithms helpful.

Grading

The final semester grade is determined as follows:

- Active participation: 25%
  - 10% - Attendance
  - 15% - Scribe
- Written problem sets: 25%
  - Problem set 1 due: 3/19/2026
  - Problem set 2 due: 4/2/2026
  - Problems are assigned during each lecture on the go
- Paper presentations: 50%
  - 10% - Propose three candidate papers and provide high-level lesson plans for each; due: 3/5/2026
  - 10% - Select one and provide a detailed lesson plan; due:4/9/2026
  - 30% - Deliver the selected paper; due: 4/9/2026 to 4/23/2026

List of papers

- Lazier then Lazy Greedy: https://arxiv.org/abs/1409.7938
- Information flow on trees: https://arxiv.org/abs/math/0107033
- Prophet Secretary: https://arxiv.org/abs/1507.01155
- Combinatorial Bernoulli Factories: https://arxiv.org/abs/2011.03865
- On coalescence time in graphs: https://arxiv.org/abs/1611.02460
- Online Bipartite Matching and AdWords: https://arxiv.org/abs/2107.10777
- Pandora's Problem with Nonobligatory Inspection: https://arxiv.org/abs/1905.01428
- Weitzman's Rule for Pandora's Box with Correlations: https://arxiv.org/abs/2301.13534
- A Knapsack Secretary Problem with Applications: https://link.springer.com/chapter/10.1007/978-3-540-74208-1_2

- Graph Sparsification by Effective Resistances https://arxiv.org/abs/0803.0929
- Random-Order Models https://www.cambridge.org/core/books/beyond-the-worstcase-analysis-of-algorithms/randomorder-models/5B094457103EFABE3BF7D85F9ABB5E1C
- Adaptivity Gaps for Stochastic Probing: Submodular and XOS Functions https://arxiv.org/abs/1608.00673
- Statistical inference on graphs: Selected Topics https://people.duke.edu/~jx77/stats-graphs.pdf (planted clique problem and graph matching problem)
- Online learning: Weighted Majority,
- Causal discovery sample complexity.
- Fastest Mixing Markov Chains
- Approximate embeddings (Johnson-Lindenstrauss Lemma, Bourgain's Theorem)
  - Isometric embeddings, tree embeddings
- Comparison theorems for the minimum eigenvalue of a random positive-semidefinite matrix https://arxiv.org/abs/2501.16578
- The power of two choices for random walks: https://arxiv.org/abs/1911.05170
  - The power of choice in random sampling https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5748962
  - Chapter 17 of Mitzenmacher, M. & Upfal, E. (Balanced Allocations and Cuckoo Hashing)
  - The Power of Two in Token Systems https://pubsonline.informs.org/doi/10.1287/mnsc.2024.06176
- Randomized Primal-Dual Analysis of RANKING for Online Bipartite Matching https://epubs.siam.org/doi/10.1137/1.9781611973105.7
- Trailing the Dovetail Shuffle to its Lair https://projecteuclid.org/journals/annals-of-applied-probability/volume-2/issue-2/Trailing-the-Dovetail-Shuffle-to-its-Lair/10.1214/aoap/1177005705.full
- Comparison theorems for the minimum eigenvalue of a random positive-semidefinite matrix https://arxiv.org/abs/2501.16578
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Reference Materials

- Mitzenmacher, M. & Upfal, E. (2017). Probability and Computing (2e).
  - http://lib.ysu.am/open_books/413311.pdf
- Harchol-Balter, M. (2023). Introduction to Probability for Computing.
  - https://www.cs.cmu.edu/~harchol/Probability/book.html
- Levin, D. A., Peres, Y., & Wilmer, E. L. (2009). Markov Chains and Mixing Times (2e).
  - https://www.cs.cmu.edu/~15859n/RelatedWork/MarkovChains-MixingTimes.pdf
- Motwani, R. & Raghavan, P. (1995). Randomized Algorithms.

Course Schedule

The course followed the schedule below (subject to adjustments for travel and student talks).

Week	Topics / Activities
Week 1, 1/15/2026	Topics: ● Tail bounds: Markov, Chebyshev, Chernoff, Hoefding, CLT Reading: ● Chapter 18 of Dr. Harchol-Balter's book
Week 2, 1/22/2026	Topics: ● Triangle of polling (revisited) ● Almost sure convergence ● Differential privacy Papers (if you want to present on differential privacy): ● McSherry & Talwar (FOCS 2007), "Mechanism Design via Differential Privacy." https://ieeexplore.ieee.org/document/4389483 ● Benefits and Pitfalls of the Exponential Mechanism with Applications to Hilbert Spaces and Functional PCA https://arxiv.org/abs/1901.10864 Problems: ● (P1) prove almost sure convergence of polls using

	Chebyshev inequality and a sandwich argument. ● (P2) Laplace mechanism for polls satisfies central differential privacy.
Week 3, 1/29/2026	● Balls and bins ● Poisson Approximation ● Random graphs Reading: Papers: Problems:
Week 4, 2/5/2026	Topics: ● Randomized algorithms ● Las Vegas algorithms ○ Expected complexity of quicksort ○ Bucket sort ● Monte Carlos Algorithms ○ Program checking (matrix multiplication, polynomial identities) ○ Randomized min-cut ● Markov chains, random walks, mixing and coupling ● Hypercube walk; s–t connectivity. Reading: Papers: Problems:
Week 5, 2/12/2026	Topics: ● Monte carlo method, 2-SAT, 3-SAT, MAX-SAT Reading:

	Papers: Problems:
Week 6, 2/19/2026	Topics: • Fully polynomial (randomized) approximation schemes • MCMC, almost uniform sampling and approximate counting Reading: Papers: Problems:
Week 7, 2/26/2026	Topics: • Probabilistic method, graph coloring, random graphs, MAX-CUT • Lovász Local Lemma Reading: Papers: Problems:
Week 8, 3/5/2026	Topics: • Sampling proper colorings; loop-erased random walks; uniform spanning trees • Martingale concentration (Azuma–Hoeffding, McDiarmid); applications (pattern matching, chromatic number). Reading: Papers:

	Problems:
Week 9, 3/12/2026	Spring break
Week 10, 3/19/2026	Topics: • Learning theory, VC dimension, PAC learning, Radermarcher complexity and bandit problems Reading: Papers: Problems:
Week 11, 3/26/2026	Topics: • Optimal stopping, secretary problem and prophet inequality Reading: Papers: Problems:
Week 12, 4/2/2026	Topics: • Maximization of monotone submodular functions ○ Knapsack, MAX-COVER • Lazier than lazy greedy Reading: Papers: Problems:

Week 13, 4/9/2026	Topics: • Paper presentations x2 Reading: Papers: Problems:
Week 14, 4/16/2026	Topics: • Paper presentations x2 Reading: Papers: Problems:
Week 15, 4/23/2026	Topics: • Paper presentations x2 Reading: Papers: Problems:

Academic Integrity

All students are expected to adhere to the standards of professional conduct and academic honesty. Collaboration for conceptual understanding is encouraged, but submitted work must be your own. University policies on academic integrity, accessibility, and inclusivity apply.

Disability Services

If you have a disability for which you may request an accommodation, contact both the instructor and Disability Resources and Services (DRS) as early as possible: 140 William Pitt Union, (412) 648-7890, drsrecep@pitt.edu.

Diversity and Inclusion

The University of Pittsburgh does not tolerate discrimination, harassment, or retaliation. Students are expected to foster a respectful learning environment. Reports can be made through the Office of Diversity and Inclusion: https://www.diversity.pitt.edu/.

Student Opinion of Teaching Surveys

Students will be asked to complete an anonymous survey near the end of the semester. Your feedback is important for improving future iterations of the course.